

Interface States in Space-Time Photonic Crystals: Topological Origin, Propagation and Amplification

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Studying the topology of spatiotemporal media poses a fundamental challenge: their remarkable properties stem from breaking spatial and temporal symmetries, yet this same breaking obscures their topological characterization. Here, we show that space-time symmetries persist in crystals with travelling-wave modulation, enabling the study of their topological properties and the prediction of spatiotemporal interface states. Using a Lorentz transformation to the frame comoving with the modulation, we identify a conserved joint parity-time-reversal symmetry in the new variables that enforces the quantization of the Zak phase, elevating it to a $\mathbb{Z}_2$ topological invariant. We then calculate the associated interface states and uncover unique features arising from time-varying effects, including selective directional excitation, propagation along moving boundaries, frequency-converted replicas, and broadband amplification even in the absence of momentum gaps.

I. INTRODUCTION

In time-varying media, the temporal modulation of a material's properties lifts the usual constraints of passive systems by breaking fundamental symmetries such as continuous time translation [1, 2]. As a result, energy conservation no longer holds, enabling light amplification and frequency conversion. When temporal and spatial modulations are combined, additional constraints are lifted, and further wave manipulation can be achieved through frequency-momentum transitions [3–5]. In particular, periodic space-time modulations of the travelling-wave form $f(x, t) = f(x - c_g t)$, where c_g corresponds to the modulation speed, have attracted significant attention since early studies [6–8]. Such modulations introduce a linear-momentum bias that enables magnet-free non-reciprocity, manifested as asymmetric band gaps supporting unidirectional propagation [9–13], as well as synthetic Fresnel drag effects in the long-wavelength limit [14]. Interestingly, such spatiotemporal (ST) phenomena originate fundamentally from wave interference effects, allowing their experimental realization across diverse platforms: from mechanical waves in water [15, 16], elastic [17, 18] and acoustic systems [19–21], to electromagnetic (EM) waves spanning a wide range of frequencies in metasurfaces [22], transmission lines [23, 24] and ENZ materials [25, 26].

Symmetries also play a central role in topological physics, where the Altland–Zirnbauer classification groups topological insulators into ten classes according to the presence or absence of three key symmetries: time-reversal, particle-hole, and chiral [27, 28]. Depending on the dimensionality of the system, each class may host topological phases identified by a characteristic bulk topological invariant. A quantized invariant predicts the

emergence of robust boundary states between materials of different phases, a principle known as bulk–boundary correspondence. These concepts can be applied to photonic crystals (PhCs), as topological invariants can also be attributed to their band structures [29–32]. However, PhCs typically lack chiral and particle-hole symmetries which, together with the modified parity of the time-reversal operator ($\mathcal{T}^2 = +1$) due to their bosonic nature, limit the number of possible non-trivial phases. Crucially, richer topological effects can emerge by invoking additional symmetries such as parity-time-duality [33, 34], which restores a fermionic-like time-reversal operator and enables the photonic analogue of the quantum spin Hall effect in photonics [35, 36]. Moreover, crystalline symmetries reveal finer topological phases within classes that would otherwise appear trivial [37–41]. For instance, one-dimensional (1D) PhCs present topological states thanks to the presence of parity (inversion) symmetry [42], with their topological invariant being the Zak phase [43].

Recently, it has been shown that the band structures of photonic time crystals, where the optical properties of an homogeneous material periodically vary in time instead of space as in PhCs, are also classified by a quantized Zak phase when the modulation preserves temporal inversion symmetry. In turn, interface states localized in time emerge [44], which can be robust against disorder for chiral-symmetric modulations [45]. Localization in time has also been proposed in different time-varying systems, as in experiments with Floquet lattices [46, 47], or two-level systems with parity-time symmetry [48].

Beyond purely temporal modulations, combining space and time variations produces ST systems with a remarkably rich topological landscape. Two broad regimes can be distinguished. First, in decoupled modulations of the form $f(x, t) = g(x)h(t)$, space and time act as independent degrees of freedom, enabling higher-dimensional topological phases such as D+1 Floquet insulators [49, 50], synthetic dimensions in frequency space

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[51–53], and systems with both spatial and temporal band gaps that allow for localization in space and time [54]. Second, in coupled ST modulations, exemplified by travelling-wave type modulations, time is no longer independent and the system remains effectively 1D, which forbids the well-established Chern classification applicable to decoupled modulations. Previous studies in acoustic systems have investigated the topology of such modulations in experiment by invoking an adiabatic approximation, which leads to a decoupled 1+1D formalism with its associated Chern number, but overlooks the effects introduced by the dynamical modulation [55]. Other works, avoiding this approximation, revealed interface states accompanied by frequency-converted replicas at frequencies far from the band gap [56]. However, in that case the topological characterization relied on a time-dependent Zak phase, leaving open the question of whether a symmetry-protected invariant underlies the existence of these states.

In this work, we answer this question affirmatively by considering a PhC whose permittivity is modulated in a travelling-wave form and analyzing the ST symmetries it supports. By performing a Lorentz transformation to the frame comoving with the modulation, we identify the conserved symmetries and show that a combined parity-time-reversal transformation remains invariant. This symmetry enforces the quantization of a *spatiotemporal* Zak phase defined along the comoving-frame Brillouin Zone (BZ), thereby establishing a $\mathbb{Z}_2$ topological invariant. Furthermore, by calculating the band energy density, we distinguish the two resulting phases given by such an invariant as a trivial phase and an obstructed atomic limit. This allows us to predict and unveil the interface states that arise between slabs of different ST Zak phases, considering both moving and static boundaries. Importantly, these states persist regardless of the modulation speed, and their resonance frequencies coincide with the bulk band-crossing position, confirming their topological origin. Owing to their ST nature, the interface states present unique features such as selective directional excitation, frequency-converted replicas and, as uncovered in this work, propagation along moving boundaries, as well as broadband amplification even in the absence of momentum gaps. Finally, we analyze the effect of relevant perturbations on the modulation, demonstrating the robustness of the interface states and clarifying the conditions under which they remain topological.

II. METHODS

Bloch-Floquet theory of travelling-wave PhCs -

We consider a medium whose permittivity is modulated in space and time following the travelling form

$$\epsilon(x, t) = \epsilon_0 \epsilon_m [1 + \alpha \cos(gx - \Omega t)], \quad (1)$$

with ϵ_m the background relative permittivity of the medium, g and Ω the spatial and temporal modulation

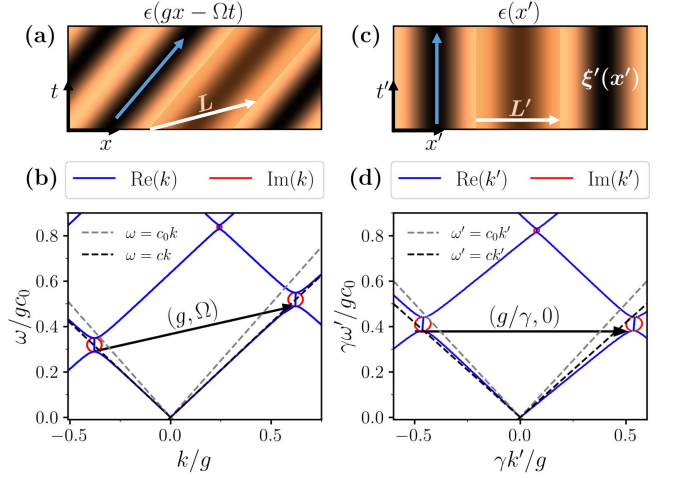


FIG. 1: Spatiotemporal photonic crystal in the laboratory and comoving frames. (a) Travelling-wave modulated permittivity in the lab-frame. A blue arrow signals the direction of the continuous space-time translation symmetry that defines the unit cell highlighted in the orange shaded region. The white arrow corresponds to the lattice vector. (b) Band structure of a STPhC for a modulation speed $c_g = \Omega/g = 0.2c_0$, modulation strength $\alpha = 0.3$ and $\epsilon_m = \mu_m = 1.2$. Grey and black dashed lines correspond to the dispersion relation in free space and in the unmodulated material, respectively. Black arrow represents the reciprocal lattice vector $\mathbf{p}$. (c) Modulated permittivity in the comoving frame. The previous magnitudes are now spatial-like, but a new magneto-electric coupling $\xi'(x')$ appears. (d) Band structure of the same STPhC in the comoving frame, with the black arrow now representing the transformed reciprocal vector $\mathbf{p}'$.

frequencies with periods $a = 2\pi/g$ and $T = 2\pi/\Omega$, and α the modulation strength. The permeability of the medium $\mu = \mu_0\mu_m$ is considered as constant, where μ_m is the background permeability, and $c_0 = 1/\sqrt{\epsilon_0\mu_0}$ is the speed of light in vacuum. As sketched in Fig. 1(a), this specific modulation where space and time are coupled creates a moving grating with phase velocity $c_g = \Omega/g$ along the x direction. Since the material itself is not moving, this grating speed can take any value up to infinity while respecting special relativity. This allows for the same model described by Eq. (1) to host different regimes depending on whether the grating velocity is lower (subluminal) or higher (superluminal) than the speed of light in the medium $c = c_0/\sqrt{\epsilon_m\mu_m}$ [14, 57], leading to drastically different behaviours. In this work, we focus on the subluminal regime, defined for our modulation as $0 \leq c_g \leq c/\sqrt{1+\alpha}$ [57], with the upper bound marking the onset of the luminal regime, which also presents unique features such as broadband unidirectional amplification [58].

We calculate the band structure of the spatiotemporal PhC (STPhC) by solving Maxwell's equations, $\nabla \times \mathbf{E} =$

$-\partial_t \mathbf{B}$ and $\nabla \times \mathbf{H} = \partial_t \mathbf{D}$. Considering s-polarization and normal incidence, the latter equations can be written in matrix form as

$$\begin{pmatrix} 0 & \partial_x \\ \partial_x & 0 \end{pmatrix} \begin{bmatrix} E_z \\ H_y \end{bmatrix} = \partial_t \left(\hat{\mathbf{M}}(x, t) \begin{bmatrix} E_z \\ H_y \end{bmatrix} \right), \quad (2)$$

where $\hat{\mathbf{M}}(x, t)$ corresponds to the constitutive matrix of the relevant components of the EM field

$$\hat{\mathbf{M}}(x, t) = \begin{pmatrix} \epsilon(x, t) & 0 \\ 0 & \mu \end{pmatrix}, \quad (3)$$

such that

$$\begin{bmatrix} D_z \\ B_y \end{bmatrix} = \hat{\mathbf{M}}(x, t) \begin{bmatrix} E_z \\ H_y \end{bmatrix}. \quad (4)$$

Then, we use the Bloch-Floquet ansatz

$$\begin{bmatrix} E_z(x, t) \\ H_y(x, t) \end{bmatrix} = e^{i(kx - \omega t)} \sum_n \begin{bmatrix} E_n \\ H_n \end{bmatrix} e^{in(gx - \Omega t)} \quad (5)$$

to derive an eigenvalue problem for the Bloch-Floquet amplitudes E_n and H_n . This allows us to compute the dispersion relation $k(\omega)$, as well as the mode decomposition of the eigenfunctions of the STPhC (see Supplemental Information (SI) for detailed derivations).

We represent the obtained band diagram of the STPhC in Fig. 1(b), together with the dispersion relation in free space and in the unmodulated medium. For our choice of parameters within the subluminal regime, the band structure displays frequency band gaps akin to spatial PhCs, in contrast to the momentum band gaps of photonic time crystals [1]. Interestingly, however, the gaps are asymmetric and appear at different frequencies for forward ($k > 0$) and backward ($k < 0$) waves, revealing the mechanism by which STPhCs enable one-way propagation [3, 4, 11]

The asymmetric band structure stems from the coupling between spatial and temporal modulations induced by the travelling wave form of Eq. (1). Since the STPhC is effectively a 1D system, it is described by a single lattice vector. While the temporal modulation breaks continuous time translation symmetry, a travelling wave modulation exhibits a continuous ST translational symmetry that conserves a linear superposition of energy and momentum [59]. This is highlighted by the blue arrow in Fig. 1(a). Such a continuous symmetry of the system defines the unit cell, sketched as a shaded trapezoid, and a ST lattice vector $\mathbf{L}$ with both spatial and temporal components. The reciprocal lattice vector is also a ST one, $\mathbf{p} = (g, \Omega)$, see Fig. 1(b), and its non-zero frequency component tilts the entire band structure and explains the asymmetry of the band gaps.

Comoving frame - To facilitate the description of the crystal's underlying ST symmetries, we reformulate our problem in a frame that comoves with the modulation

[57, 60–65]. After a Lorentz boost in the x direction, the comoving coordinates read

$$x' = \gamma(x - c_g t), \quad y' = y, \quad z' = z, \quad (6)$$

and

$$t' = \gamma\left(t - \frac{c_g}{c_0^2}x\right), \quad (7)$$

where the Lorentz factor $\gamma = 1/\sqrt{1 - (c_g/c_0)^2}$. By transforming the EM fields accordingly [66], we then obtain Maxwell's equations in the comoving frame

$$\begin{pmatrix} 0 & \partial_{x'} \\ \partial_{x'} & 0 \end{pmatrix} \begin{bmatrix} E'_z \\ H'_y \end{bmatrix} = \hat{\mathbf{M}}'(x') \partial_{t'} \begin{bmatrix} E'_z \\ H'_y \end{bmatrix}, \quad (8)$$

with the transformed constitutive matrix

$$\hat{\mathbf{M}}'(x') = \begin{pmatrix} \epsilon'_\perp(x') & \xi'(x') \\ \xi'^T(x') & \mu'_\perp(x') \end{pmatrix}, \quad (9)$$

whose elements in the new frame read

$$\epsilon'_\perp(x') = \frac{\epsilon(x')}{\gamma^2(1 - \epsilon(x')\mu c_g^2)}, \quad (10)$$

$$\mu'_\perp(x') = \frac{\mu}{\gamma^2(1 - \epsilon(x')\mu c_g^2)}, \quad (11)$$

and

$$\xi'(x') = c_g \frac{\epsilon(x')\mu - c_0^{-2}}{1 - \epsilon(x')\mu c_g^2}. \quad (12)$$

In the comoving frame, the constitutive parameters depend on the spatial coordinate x' only, and the modulation loses the explicit temporal dependence. More importantly, moving frames leads the material to acquire a bianisotropic coupling $\xi'(x')$ proportional to the grating velocity, resulting in a moving-medium type coupling between the electric and magnetic fields [57].

We sketch the permittivity distribution in the comoving frame in Fig. 1(c). In this frame, there is a continuous ST translation symmetry along t' , such that the unit cell is defined with the spatial coordinate only as $x' \in [-\gamma a/2, \gamma a/2]$, with lattice vector $\mathbf{L}' = (\gamma a, 0)$. Using the reciprocal coordinates in the comoving frame

$$k' = \gamma\left(k - \frac{c_g}{c_0^2}\omega\right), \quad k'_y = k_y, \quad k'_z = k_z, \quad (13)$$

and

$$\omega' = \gamma(\omega - c_g k), \quad (14)$$

we obtain the reciprocal lattice vector $\mathbf{p}' = (g/\gamma, 0)$ as well as the BZ defined as $k' \in [-g/2\gamma, g/2\gamma]$. Finally, transforming back to the laboratory frame, the corresponding lattice vector can be calculated as $\mathbf{L} = \gamma^2 a(1, c_g/c_0^2)$.

With the lattice vector now established in the comoving frame, the periodicity of the crystal becomes explicit, namely, $\tilde{\mathbf{M}}'(x' + \gamma a) = \tilde{\mathbf{M}}'(x')$. This allows us to apply Bloch's theorem to Eq. (8) and obtain the dispersion relation $\omega'(k')$ together with the comoving-frame eigenfunctions

$$\begin{bmatrix} E'(x') \\ H'(x') \end{bmatrix}_{k',m} = e^{ik'x'} \begin{bmatrix} u'^E(x') \\ u'^H(x') \end{bmatrix}_{k',m}, \quad (15)$$

where the subscripts of the electric and magnetic fields have been dropped for clarity, $u'^{E/H}_{k',m}(x' + \gamma a) = u'^{E/H}_{k',m}(x')$ correspond to the periodic part of the eigenfields, and m represents the band index (see SI).

Figure 1(d) shows the dispersion relation of the STPhC in the comoving frame. The absence of frequency component in $\mathbf{p}'$ removes the tilted nature of the band structure observed in the laboratory frame [cf. Fig. 1(b)], rendering the band gaps symmetric. However, the system still presents non-reciprocal features, as visible from the fact that $\omega'(-k') \neq \omega'(k')$. Indeed, an increased group velocity is observed for backward propagating waves, while forward propagating waves present a reduced one. This is a result of the non-zero magneto-electric coupling that appears in the comoving frame, which breaks time-reversal symmetry $\mathcal{T}'(t' \rightarrow -t')$ and, consequently, reciprocity [67].

Therefore, although the reciprocal lattice vector is spatial-like in the comoving frame, the effective magneto-electric coupling induced by the temporal modulation still underlies the breaking of fundamental symmetries as well as the emergence of non-reciprocal features. In the following, we analyze the impact of this modulation-induced bianisotropic coupling on the ST symmetries present in the comoving frame, as well as its consequences in the topological characterization.

III. RESULTS

Topological characterization of STPhCs - In spatial non-magnetic PhCs, only time-reversal $\mathcal{T}$ is present out of the three fundamental symmetries that define the ten-fold way [27, 28], which for general 1D systems classify them as trivial. However, crystalline symmetries can enrich PhCs with nontrivial topology [38]. In 1D, this role is played by parity (inversion) symmetry $\mathcal{P}$ ($x \rightarrow -x$), which quantizes the Zak phase [43], defining it as a $\mathbb{Z}_2$ topological invariant. This symmetry also establishes a direct connection between the parity of the Bloch functions and the existence of surface states [42, 68], i.e., it establishes a bulk-interface correspondence.

Motivated by this context, we now discuss the topological characterization of a STPhC by studying the space-time counterpart of parity symmetry, defined in the comoving frame as $\mathcal{P}'(x' \rightarrow -x')$. Applying this transformation to Eq. (8) reveals that, although each entry of the material matrix is symmetric under $\mathcal{P}'$, the

presence of the magneto-electric coupling ξ' breaks the invariance of Maxwell's equations. As a result, the direct generalization of the symmetry that enables non-trivial topology in static crystals is not possible, raising the question of whether STPhCs can still be topologically classified. Crucially, however, while $\mathcal{T}'$ and $\mathcal{P}'$ are individually broken, their combined operation $\mathcal{P}'\mathcal{T}'[(x', t') \rightarrow -(x', t')]$ is preserved. In what follows, we examine whether this joint symmetry can play a similar role as parity symmetry does in 1D spatial PhCs.

To do so, we study the Zak phase defined with the eigenfunctions of Eq. (15) and along the comoving-frame BZ

$$\theta_m^{\text{ST}} = i \int_{-g/2\gamma}^{+g/2\gamma} dk' \langle \mathbf{u}'_{k',m} | \partial_{k'} \mathbf{u}'_{k',m} \rangle, \quad (16)$$

where the integrand corresponds to the Berry connection and $\mathbf{u}'_{k',m} = [u'^E_{k',m}, u'^H_{k',m}]^T$. Importantly, the Hermiticity of the constitutive matrix in Eq. (9), together with $\mathcal{P}'\mathcal{T}'$ symmetry, allows us to relate the left and right eigenvectors of the system (see SI). As a result, the conventional Berry connection can be used in place of the biorthogonal one, which would otherwise be required by the non-Hermitian nature of the differential operator in Eq. (8). This Berry connection is defined with a weighted scalar product over the unit cell given by

$$\langle \Psi'_1 | \Psi'_2 \rangle = \int_{-\gamma a/2}^{+\gamma a/2} dx' \Psi'^{\dagger}_1(x') \hat{\mathbf{M}}'(x') \Psi'_2(x'). \quad (17)$$

The scalar product ensures the normalization of the comoving-frame eigenfunctions by establishing $\langle \Psi'_{k',m} | \Psi'_{k',m} \rangle = 1$, with $\Psi'_{k',m} = [E'_{k',m}, H'_{k',m}]^T$.

Furthermore, building upon the results of Ref. [69] on the Hermitian Berry connection, one can show that $\mathcal{P}'\mathcal{T}'$ -symmetry alone allows the quantization of the ST Zak phase. The quantization defines θ_m^{ST} as a $\mathbb{Z}_2$ topological invariant, which means that one can distinguish between two different phases, delimited by a topological transition where the band gap must close. Therefore, solely from this property we can predict the existence of interface states between two STPhCs of different topological phase.

To understand the physical meaning behind these two phases, we use Eq. (17) to establish a direct relation between Wannier centers and the ST Zak phase, as is done in the Hermitian case [70]. We can then write

$$\bar{x}'_m = \langle \mathbf{w}'_{L',m} | x' | \mathbf{w}'_{L',m} \rangle = \gamma \theta_m^{\text{ST}} / g, \quad (18)$$

with $\mathbf{w}'_{L',m}(x') = \gamma a / 2\pi \int dk' e^{ik'L'} \Psi'_{k',m}$ the Wannier function associated to the eigenfields $\Psi'_{k',m}$ and $L' = n\gamma a$, with $n \in \mathbb{Z}$, labeling the position of the unit cell. Therefore, the quantization of θ_m^{ST} enforced by $\mathcal{P}'\mathcal{T}'$ -symmetry translates directly into the quantization of the Wannier centers. In particular, since the Zak phase can only take the values $0, \pi \pmod{2\pi}$, the Wannier functions

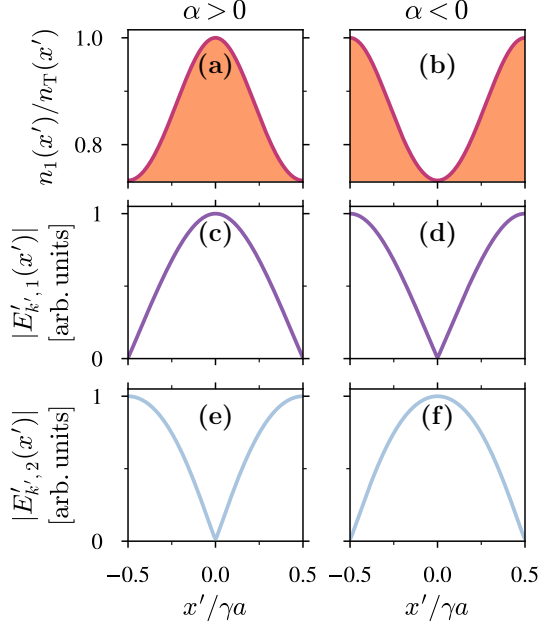


FIG. 2: Distinct topological phases in STPhCs. (a)-(b) Energy density distribution over the unit cell of the first band $m = 1$, for $\alpha > 0$ and $\alpha < 0$. (c)-(d) Absolute value of the electric field eigenfunction $|E'_{k',m=1}(x')|$ located at the first band gap shown in Fig. 1(d) for $k' < 0$, considering $\alpha > 0$ and $\alpha < 0$. (e)-(f) $|E'_{k',m=2}(x')|$ at the same gap for the second band $m = 2$, considering $\alpha > 0$ and $\alpha < 0$. All the parameters are the same as in Fig. 1.

can only be localized at the positions $\bar{x}' = 0, \gamma a/2$, which correspond to the inversion centers of the unit cell.

In electronic insulators, Wannier centers mark the positions of electronic charge within the unit cell, effectively defining atomic orbitals [70]. The quantization of $\bar{x}'_m$ thus distinguishes two phases, with orbitals localized at the cell center or edges. Within the band representation framework [38, 71, 72], these correspond in one dimension to trivial and obstructed (topological) phases, respectively. This formalism extends to PhCs, where EM energy density acts as an analogue of electronic charge [73], enabling the same physical interpretation of the two phases defined by the ST Zak phase in our photonic system. Indeed, this can be derived directly from the time-averaged energy density of the comoving-frame eigenfields. Integrating over the whole BZ, the latter density reads [66]

$$n_m(x') = \int_{-g/2\gamma}^{+g/2\gamma} \frac{E'_{k',m} D'^*_{k',m} + H'_{k',m} B'^*_{k',m}}{2} dk', \quad (19)$$

where $D'_{k',m}$ and $B'_{k',m}$ correspond to the displacement and magnetic flux eigenfields, respectively, obtained through the constitutive matrix in Eq. (9). The total EM energy density is then $n_T(x') = \sum_m n_m(x')$.

In order to corroborate the existence of the two topologically distinct phases in our model, we study the transition between positive and negative modulation strength, with $\alpha = 0$ in Eq. (1) marking the topological transition point. This is shown in Fig. 2(a)–(b), where we represent the distribution of the energy density of the first band $m = 1$ along the unit cell for both phases. Indeed, we find that for $\alpha > 0$ the energy density of Eq. (19) is localized at the center of the unit cell, corresponding to the trivial phase, whereas for $\alpha < 0$ it shifts to the edges, which marks the obstructed (topological) phase [see Fig. 2(b)]. We note that a subtlety arises in the calculation of θ_m^{ST} for the first band $m = 1$ near the singular point $k' = \omega' = 0$, where the parity of the electric and magnetic fields becomes ill-defined. Even though this does not affect the quantization of the Zak phase, it obscures the direct correspondence with the Wannier centers (see SI). Crucially, only the relative difference of Zak phases between crystals matters for the appearance of interface states [74, 75], so this subtlety does not affect the topological characterization of STPhCs.

In Fig. 2(c)–(f) we further represent the electric eigenfields at the gap between the first and second bands for both signs of α . The eigenfunctions clearly interchange across the $\alpha = 0$ transition, a hallmark of band inversion that, as established in Ref. [76], characterizes a topological phase transition even in non-Hermitian systems as long as they remain in the $\mathcal{P}'\mathcal{T}'$ -exact phase. Taken together, these results confirm that $\mathcal{P}'\mathcal{T}'$ -symmetry alone suffices to topologically classify STPhCs, ensuring the existence of two distinct phases even in the absence of parity symmetry.

Interface state - Building on the prediction of two topologically distinct phases, we now employ a semi-analytical scattering matrix formalism (detailed in the SI) to investigate the presence of interface states at the boundary between two STPhCs, as well as discuss their novel properties due to the temporal modulation. To do so, we consider two types of slabs by truncating the same STPhC in different space-time directions: along the modulation (ST boundary) [see Fig. 3(a)] and at a fixed position in space (spatial boundary) [see Fig. 3(f)].

Spatiotemporal Interface - We begin by considering two ST slabs, each composed of seven unit cells of the STPhC defined in Eq. (1), as represented in Fig. 3(a). Specifically, we set the medium's parameters to $\epsilon_m = \mu_m = 1.2$ and $\alpha = 0.3$ as in Figs. 1 and 2, with different signs of the modulation strength α for each slab. As discussed in the previous section, positive and negative values of α correspond to different topological phases, and consequently, interface states are expected to appear. In Fig. 3(b), we show the transmission spectrum of this configuration as a function of the grating speed c_g , obtained by exciting the left slab with a plane wave and measuring the transmitted wave at the right end. Horizontal cross sections at $c_g = 0.3c_0$ and $c_g = 0.57c_0$ are presented in Fig. 3(c)–(d), respectively. We observe that the darker regions

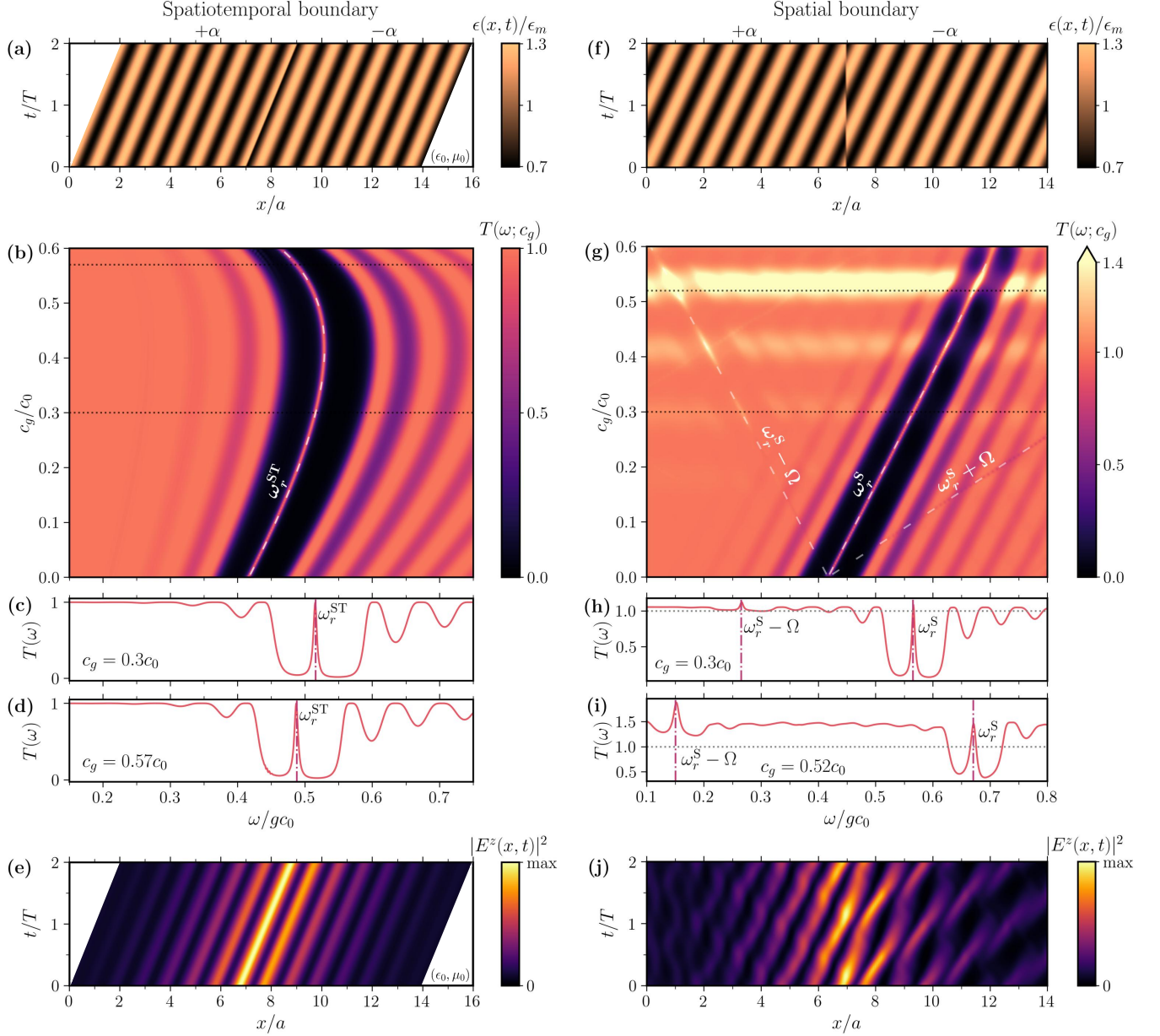


FIG. 3: Interface states between a trivial ($\alpha > 0$) and a topological ($\alpha < 0$) spatiotemporal slab, considering (a)-(e) a spatiotemporal boundary and (f)-(j) a spatial boundary. (a) Permittivity profile $\epsilon(x, t)$ of the two-slab configuration for $c_g = 0.3c_0$. (b) Transmittance spectrum of two spatiotemporal slabs whose boundary moves along with the modulation, as a function of the modulation speed c_g . An interface state with resonance frequency following the bulk-predicted ω_r^{ST} is observed. (c) Horizontal cross section of the transmittance at $c_g = 0.3c_0$. A clear peak inside the gap corresponding to the interface state is visible. (d) Same magnitude for $c_g = 0.57c_0$. (e) Intensity distribution of the interface state observed when exciting the left slab with a plane wave of frequency ω_r^{ST} . A propagating interface state is observed. (f) Permittivity profile $\epsilon(x, t)$ of the two spatial slabs for $c_g = 0.3c_0$. (g) Transmittance spectrum as a function of c_g for two slabs of the STPhC with purely spatial boundaries. An interface state is also present for the whole modulation interval. Furthermore, the frequency conversion induced by the temporal modulation enables replicas of the interface states at $\omega_r^S \pm \Omega$. (h) Horizontal cross section of the transmittance for $c_g = 0.3c_0$, revealing the interface state inside the gap as well as the replicas at $\omega_r^S - \Omega$. (i) Same magnitude for $c_g = 0.52c_0$, where broadband amplification ($T(\omega) > 1$) is observed. (j) Intensity distribution of the interface state for a static boundary, showing no propagation.

in Fig. 3(a), corresponding to lower transmission, mark the band gap, while the continuous peak within the gap identifies the interface state. Its persistence across the entire modulation range confirms the prediction of the ST Zak phase and supports the conclusion that $\mathcal{P}'\mathcal{T}'$ -symmetry alone suffices for the topological characterization of the system. Indeed, despite the progressive breaking of $\mathcal{P}'$ -symmetry with increasing magneto-electric coupling $\xi'(x'; c_g)$, the interface state persists.

Furthermore, we can predict the resonance frequency ω_r^{ST} at which interface states will be excited for each c_g through the following intuition: a topological transition necessarily entails a band-gap closing, so the frequency at which this occurs identifies the precise point in the band structure where an interface state must emerge between materials at opposite sides of this transition. Therefore, the resonance frequency can be determined by finding the band-crossing position in the unmodulated case $\alpha = 0$, which marks the transition point.

In order to find the band crossing point that we argue predicts the resonance frequency of the interface state, we use homogenization theory to obtain a linear approximation of the band dispersion of the STPhC. The dispersion is constructed by copying replicas of the $\omega = \pm v_{\text{eff}}k$ curves, where $v_{\text{eff}} = 1/\sqrt{\epsilon_{\text{eff}}\mu_{\text{eff}}}$ corresponds to the effective velocity defined by the homogenized material parameters (see Ref. [57]). These replicas are translated by the reciprocal lattice vector $\mathbf{p} = (g, \Omega)$ an integer number of times n , yielding $\omega + n\Omega = \pm v_{\text{eff}}(k + ng)$. Then, calculating the crossing point between the fundamental ($n = 0$) forward mode, and the first replica of the backward mode ($n = -1$), we obtain the crossing point corresponding to the first gap in the forward direction (see sketch in SI):

$$\frac{\omega_{\text{gap}}}{gc_0} = \frac{c_g + v_{\text{eff}}(c_g)}{2c_0}. \quad (20)$$

However, since we are considering a ST interface, it is essential to account for the modified conservation law: it is ω' (the comoving frequency) that is conserved, rather than ω . This means that, when a monochromatic wave of frequency ω_0 encounters the first interface coming from free space, it will not couple to the same frequency inside the material ω_{mat} , but rather to a Doppler shifted frequency given by the expression [60]

$$\omega'/\gamma = (1 - c_g/v_{\text{eff}})\omega_{\text{mat}} = (1 - c_g/c_0)\omega_0. \quad (21)$$

This explains why the band-gap position in Fig. 3(b) does not grow linearly with c_g , as might be expected from the tilt induced by the reciprocal lattice vector, but instead shifts to lower frequencies once the modulation speed becomes sufficiently large. This red shift effect can be clearly seen by comparing the transmission spectra for $c_g = 0.3c_0$ [Fig. 3(c)] and $c_g = 0.57c_0$ [Fig. 3(d)]. As expressed in Eq. (21), when $c_0 \neq v_{\text{eff}}$ the mismatch between ω_0 and ω_{mat} grows with increasing c_g , leading to the more intricate dependence observed in Fig. 3(b). Interestingly, this effect makes the gap felt by the incoming wave larger than the width of the actual band gap.

Furthermore, from Eq. (21) we can directly determine the incident frequency necessary to excite ω_{gap} inside the material, which corresponds to the resonance frequency of the topological states at a ST interface:

$$\omega_r^{\text{ST}} = \frac{1 - c_g/v_{\text{eff}}}{1 - c_g/c_0} \omega_{\text{gap}}. \quad (22)$$

This frequency is shown as a function of c_g in Fig. 3(b) with a dashed line and, as we can see, it predicts with high accuracy the numerically calculated position of the interface state within the gap, showing the connection of these states to the properties of the bulk. This connection, together with the persistence of the interface states, establishes a ST analogue of the bulk-interface correspondence of 1D spatial PhCs.

Finally, we present in Fig. 3(e) the intensity distribution of the interface state excited by a plane wave at its resonance frequency ω_r^{ST} considering a grating velocity $c_g = 0.3c_0$. As observed, the topological state remains localized, decaying evanescently away from the moving boundary. Interestingly, this creates a 1D propagating interface state, in contrast to the 0D static state typically found in 1D spatial PhCs. This behaviour stems from the fact that, even if the infinite crystal is effectively 1D, ST boundaries confer on the system a phenomenology that goes beyond that of purely 1D crystals.

Spatial interface - We now turn to the case of two spatial slabs of the same STPhC, as illustrated in Fig. 3(f), in order to examine how changing the type of boundary affects the interface state. The static interface implies that the slabs are constructed with a purely spatial unit cell, defined with the same spatial period as the modulation $a = 2\pi/g$. Since this does not correspond to the ST unit cell used to characterize the infinite crystal, the derivation of the scattering matrix for this case needs a tailored approach (see SI). As for the ST interface, we plot in Fig. 3(g) the transmittance spectrum of the slabs in this configuration as a function of the modulation velocity, together with two horizontal cross sections at $c_g = 0.3c_0$ and $c_g = 0.52c_0$. Crucially, we find that an interface state also appears inside the gap regardless of the value of the grating speed. This demonstrates the robustness and predictive power of the ST Zak phase, which correctly anticipates the existence of such states for two very different types of boundaries. Furthermore, since the spatial interface conserves the lab-frame frequency ω , we can directly predict the resonance frequency of this state through Eq. (20) as $\omega_r^{\text{S}} = \omega_{\text{gap}}$, without the need of any Doppler shift correction as for the ST boundary case.

From Fig. 3(g) we observe two key differences from the ST boundary case: (i) the appearance of replicas of the interface state at frequencies that lie outside the band gap, highlighted by dashed lines, and (ii) the amplification of the transmitted wave with $T(\omega) > 1$.

First, as reported in Ref. [56] for a ST phononic crystal, the temporal modulation enables the excitation of the interface state from frequencies shifted in multiples of Ω

thanks to the frequency conversion processes occurring at each boundary. This is confirmed by the appearance of a second-order transmission peak at $\omega_r^S - \Omega$ in Fig. 3(h)-(i) highlighted by a vertical line. To understand why this frequency conversion only happens for spatial slabs, we again put our attention on the conservation law for each boundary: conservation of the incident lab-frame frequency ω implies that an incoming plane wave excites all the modes of the band structure that lay in the horizontal line defined by a given ω_0 , whereas conservation of ω' implies exciting all the modes laying in a diagonal line with slope equal to c_g [60]. As a consequence, once we fold each mode into the BZ by shifting them an integer number of reciprocal vectors $\mathbf{p}$, we find that the spatial boundary excites an infinite number of modes inside the BZ with frequencies that differ an integer number of Ω with respect to ω_0 [11] (see sketch in SI). However, for the ST boundary, each mode along the diagonal line folds back into the same mode inside the BZ, such that we can only excite one truly distinct mode inside the material, explaining why we do not observe frequency conversion in this configuration. We have verified this by means of our semi-analytical scattering matrix formalism, which allows us to study the mode decomposition of the scattered fields, as well as the fields inside the slabs (see SI).

Second, the amplification of the transmitted wave can be clearly observed by the horizontal bands of transmittance larger than one in Fig. 3(g), together with Fig. 3(i) where most part of the transmission spectrum present $T(\omega) > 1$, highlighting the broadband nature of this effect. Within our framework, we can identify frequency conversion processes as the underlying mechanism behind the amplification. Indeed, as shown in the SI, the permittivity encountered by the wave at the second and third interfaces determines whether higher-frequency modes are created in the scattering processes. For a given size of the slab, this condition is satisfied depending on the modulation velocity, explaining the periodic character of the amplification. Moreover, the amplification increases with c_g , since the wave experiences more modulation cycles while traversing the finite slab. In contrast, this effect is absent at ST boundaries, where frequency conversion does not occur. Crucially, this mechanism enables amplification even in the subluminal regime, without the need for the opening of a momentum gap as in photonic time crystal [77].

Finally, we plot in Fig. 3(j) the interface state excited by an incoming plane wave of frequency ω_r^S considering a modulation velocity $c_g = 0.3c_0$. As we can see, although the overall intensity profile presents a similar spatial structure to Fig. 3(e), the peak of the field's intensity remains pinned at the static interface, and the state does not propagate without perturbation along the boundary, in contrast to the ST boundary case.

Robustness against perturbations - An important aspect of interface states of topological origin is their inherent robustness against certain perturbations. In

the previous section, we showed that the interface state presents robustness against an increase of the symmetry-breaking parameter ξ' . Indeed, not only did the states persist, but their resonance frequency remained pinned to the one predicted by the band-crossing position $\omega_r^{ST/S}$. Here, we extend this analysis by modifying the properties of one of the slabs, such that the two are no longer related by a simple spatial shift. Moreover, we also discuss the effect of truncating the crystal at arbitrary points rather than at the inversion centers. These configurations represent more general and realistic scenarios, allowing us to test whether the robustness of the interface states extends beyond idealized conditions that might not be met in experimental realizations.

Specifically, we revisit the configuration of ST slabs of Fig. 3(a), introducing a perturbation Δ that increases the modulation strength of the second slab to $\alpha + \Delta$. In Fig. 4(a), we evaluate numerically the resonance frequency ω_i of the interface state and plot it as a function of Δ and c_g to study its deviation from the bulk prediction ω_r^{ST} . Importantly, while this frequency plays a role similar to the mid-gap position in chiral-symmetric systems, here a deviation does not always signal a loss of bulk connection, and thus the source of the deviation needs to be carefully studied. In the standard case of a spatial 1D PhC, $c_g = 0$, we observe no deviation with Δ , highlighting the robustness of the interface states, since they are completely insensitive to the perturbation. As we turn on and increase the modulation speed, however, the asymmetry in the properties of each slab becomes more relevant, as the deviation grows with increasing Δ and c_g . This effect is clearly observed when comparing the position of the transmission peaks for $c_g = 0.1c_0$ and $c_g = 0.5c_0$ in Fig. 4(b)-(c), respectively.

The deviation can be understood by noting that the effective parameters ($\epsilon_{\text{eff}}, \mu_{\text{eff}}$) used to derive the band-crossing position depend on the modulation strength when $c_g \neq 0$ [57]. Indeed, Fig. 4(d) shows how the predicted resonance frequency ω_r^{ST} for a STPhC with modulation strength $\alpha + \Delta$ splits into distinct values for different Δ as the modulation speed increases. Consequently, in a finite system composed of two slabs with different modulation strengths, each slab predicts a distinct ω_r^{ST} , explaining the deviation and the apparent loss of predictive power. Crucially, however, this does not imply a breakdown of bulk-interface correspondence, but rather highlights the need for a more refined bulk theory to determine the exact band-crossing position in such asymmetric slab configurations. To emphasize this point, we note that the deviation in frequency observed in Fig. 4 is therefore different in nature from the one that would induce a phase perturbation ϕ in the modulation Eq. (1). Indeed, such a shift truncates the slabs at points different from the inversion centers of the unit cells, which obscures the distinction between trivial and topological phases by lifting the quantization of the Zak phase. While interface states may still appear in that configuration, as reported for the static case in Refs. [78, 79],

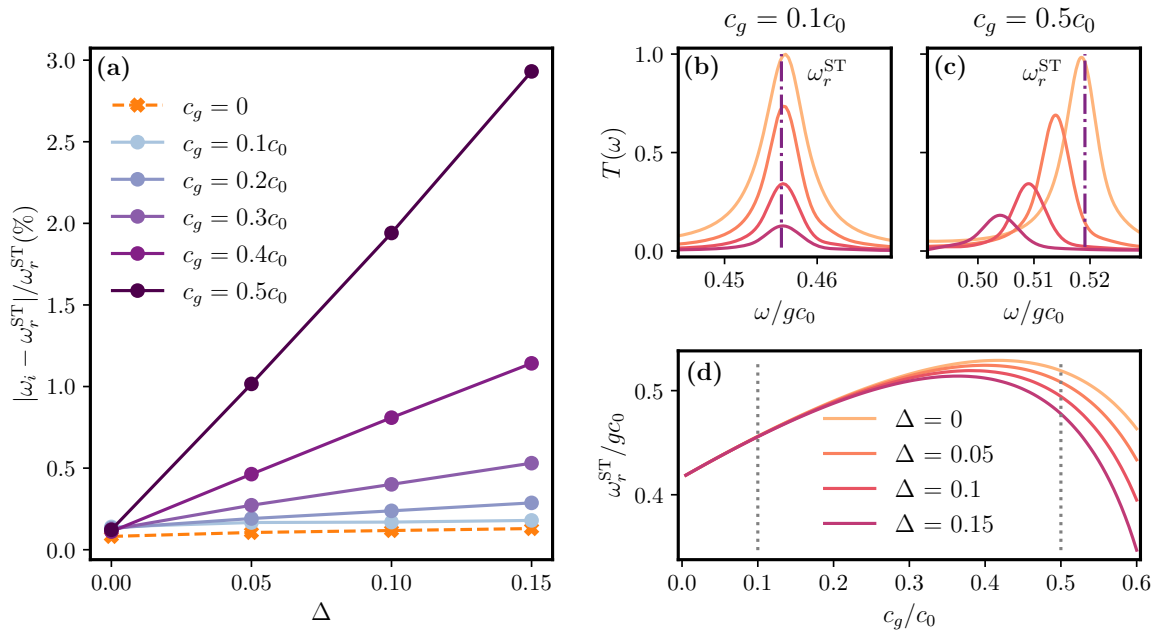


FIG. 4: Robustness of interface states studied in the two-slab configuration of Fig. 3(a) by perturbing the system with an increased modulation strength in the second slab by $(\alpha + \Delta)$, with Δ the perturbation parameter. (a) Deviation of the numerically obtained resonance frequency ω_i from the bulk-predicted ω_r^{ST} as we increase Δ and c_g . (b) Zoom in at the peaks of transmittance observed inside the gap for $c_g = 0.1c_0$, with a vertical line showing ω_r^{ST} as a reference. (c) Same magnitude for $c_g = 0.5c_0$. (d) Analytical resonance frequency ω_r^{ST} for different modulation strengths $(\alpha + \Delta)$ as a function of the modulation speed c_g .

their resonance frequency cannot be related anymore to the bulk-predicted value, signaling in that case a loss of topological nature.

IV. CONCLUSIONS

In this work, we investigate the topological origin of interface states in spatiotemporal photonic crystals with travelling-wave modulation and uncover their unique properties. Through the appropriate consideration of the symmetries present in the system, we provide a framework to topologically classify time-dependent modulations of travelling-wave type. Notably, we show that computing topological invariants in a frame that co-moves with the modulation leads to a well-defined and meaningful topological classification. Making use of a Lorentz transformation to the frame that moves along with the modulation, we show that the spatiotemporal counterpart of parity-time-reversal symmetry is conserved. This symmetry enforces the quantization of the spatiotemporal Zak phase defined along the comoving-frame Brillouin Zone, yielding it a $\mathbb{Z}_2$ topological invariant. By calculating the electromagnetic band energy density, we distinguish the two resulting phases as a trivial phase and an obstructed atomic limit, which clarifies the topological nature of interface states in spatiotemporal photonic

crystals without further symmetries.

We prove this point by calculating semi-analytically the interface states that arise between slabs of different spatiotemporal Zak phase for two types of boundaries: spatiotemporal and purely spatial. From the transmittance spectra, we find that an in-gap state exists throughout the entire subluminal regime in both configurations, and that its resonance frequency can be accurately predicted from bulk properties. The interface states also present novel features stemming from the non-reciprocal nature of travelling-wave media and the lack of energy conservation. For a spatiotemporal interface, the topological state propagates along with the moving boundary, a remarkable feature in an effectively one-dimensional system. In contrast, a static interface introduces frequency conversion at each scattering event, producing replicas of the interface states and broadband amplification of the transmitted wave even without momentum gaps. Furthermore, non-reciprocity in this system also enables selective excitation: the topological state can be accessed from one propagation direction while the opposite direction shows full transmission.

Our results highlight the potential of temporal modulation to enrich the properties of interface states in spatiotemporal crystals. This paves the way to explore time-varying effects in higher-dimensional systems with additional symmetries, where richer topological phases

already exist in the static limit, or within the same model considered here by extending our framework to the superluminal regime. Although this work focuses on photonic systems, our predictions are general and extend to other wave platforms where spatiotemporal interfaces and travelling-wave modulations have already been experimentally demonstrated: elastic materials such as

elastic strips [17] or piezoelectric crystals [18], acoustic systems [55], and transmission lines [24]. This underscores the feasibility of realizing these effects experimentally and broadens their potential applications beyond those of their static counterparts [80–82]. Altogether, these directions point to a broader landscape in which spatiotemporal modulation becomes a key ingredient for engineering novel topological phases for wave control.

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